

Temporal Provider-Side Fairness in Information Retrieval: An Axiomatic Perspective

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Abstract

Provider fairness in information retrieval (IR) often demands fair representation of both individual providers and groups of providers in the retrieval output. We formulate axioms for both types, inspired by similar settings in social choice theory, with the aim of long-term (amortised) fairness across queries.

For individual fairness, we draw on notions of envy-freeness and proportionality from work on fair division. Intersectional group fairness can be expressed as a multidimensional apportionment problem. In contrast to classical social choice settings, we introduce relevance as a key parameter for these axioms. While these two types of fairness are in general incompatible, we discuss weakened axioms and simplifying assumptions in order to obtain simple model settings where the two can be unified.

1 Summary

In information retrieval (IR), the goal is to retrieve a small number of highly relevant items given a user or query as input. Popular applications of IR include recommender systems (RS), which predict the most relevant items (for example songs or movies) for a particular user, and retrieval-augmented generation (RAG) systems, which retrieve information relevant to an LLM query and use it to inform the generation of a response.

A main concern in IR is the quality of retrievals. This can be measured in different ways, e.g., using a test data set to validate relevance predictions with empirical data [Järvelin and Kekäläinen, 2000], or by ensuring proportional representation of the dataset in the retrievals [Steck, 2018], referred to as *calibration* in RS literature (where it commonly used as a measure of personalisation for each individual user).

In many IR settings, fairness plays a crucial role: users should experience a similar retrieval quality, providers of items should receive a fair share of exposure and should not be disproportionately (dis)advantaged for reasons like gender or popularity. Providers of items may be institutions issuing information about their offers, news outlets, companies selling products or artists authoring songs. Additionally, external

fairness constraints may need to be enforced, such quotas of representation for certain groups of item providers.

A typical taxonomy of the various fairness types in IR distinguishes along three axes: temporal (short-term vs long-term), scope (individual vs group), and objective (user- vs provider-side) [Deldjoo *et al.*, 2024]. We focus on individual and group provider-side fairness. That is, we are concerned with the fair allocation of exposure for single providers of information (e.g. artists in RS or authors in RAG), and the distribution of exposure across (e.g. demographic or opinion-based) groups of such providers. Because many IR-based systems operate continuously, influencing information exposure or decision-making at a collective level, we study long-term fairness.

Our attention to long-term fairness is not concerned with the dynamic behaviour and development of the IR system over time (e.g. through feedback loops) [Ekstrand *et al.*, 2022, Chapter 7]. Instead, we focus on temporality as an additional dimension which allows us to allocate exposure fairly *across queries and users*. This is similar in spirit to the idea of perpetual voting or temporal fair division in computational social choice [Lackner, 2020; Elkind *et al.*, 2024b; Elkind *et al.*, 2024a]: Because fairness goals cannot be satisfied in each allocation round, we allow their violation in individual rounds and instead balance them over time. In the IR literature, this angle is sometimes discussed as *amortised* fairness [Biega *et al.*, 2018; Xu *et al.*, 2024]. For a RS application, a similar approach using social choice functions for fairness-aware re-ranking of RS rankings has been developed [Aird *et al.*, 2025].

We additionally distinguish along the axis of exposure-based vs. relevance-based fairness [Deldjoo *et al.*, 2024; Balagopalan *et al.*, 2023]. This depends on the interpretation of the IR output. If users of a recommender system, for example, can be assumed to simply consume the first few recommended items, exposure-based fairness is more relevant. On the other hand, if they only pick the most relevant of the recommended items – or none at all if their relevance is too low –, relevance-based fairness is central. For the formulation of axioms, this distinction determines whether the allocation of exposure should depend on the relevance of the retrieved items or not.

Setting. We have a number of *providers* providing pairwise disjoint sets of items. The aim of the IR system is to retrieve

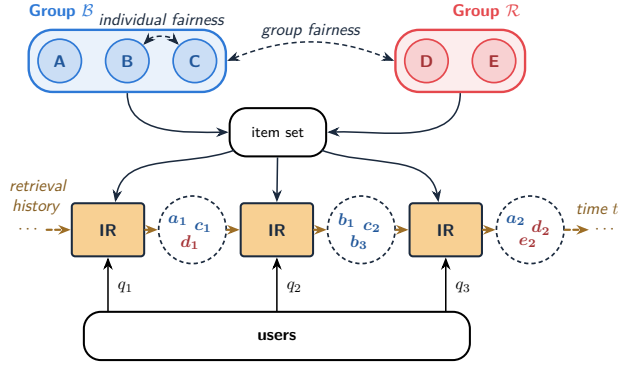


Figure 1: An exemplary scenario with two provider groups $\mathcal{B}$ and $\mathcal{R}$, each consisting of several providers. At each time step t_i , the IR mechanism takes a user query q_i (and a corresponding relevance score for the items), the history of retrieval sets, and the item set of all providers to output a set of relevant items. Both individual and group fairness refer to the set of all retrievals across all queries.

items from the union of all of these sets – the *item set*. The set of providers can be partitioned into groups according to various properties (gender, language, etc.). For each property, we have a target distribution specifying the target fractions of representation (in terms of exposure) for each group.

In each round, a user arrives with a *query*. A *relevance function* then assigns a score to each item in the item set that depends on the query. An *IR mechanism* takes as inputs the items and their relevance values, the current user and query, as well as the history of retrievals up to the current round, and returns a fixed-size *retrieval set* of items.

We connect individual and group provider fairness in this setting to established settings from social choice theory, namely fair division and proportional apportionment, respectively. Additional complexity comes in through the online nature of the IR setting: Queries and therefore per-round relevance scores for the items cannot be known in advance. A schematic overview of our setting is shown in Figure 1.

Individual Provider Fairness. For individual provider fairness, we aim to guarantee each provider a fair share of exposure across all rounds. This resembles temporal fair division of indivisible items [Elkind *et al.*, 2024a], where the spots in the retrieval set are allocated to the different providers (resp. their items). Fairness in this setting is typically phrased as envy-freeness or proportionality.

Envy-freeness is based on agent-to-agent comparisons: In the IR setting, no provider should prefer another provider’s allocated exposure to their own exposure. As it stands, this would be too strong: We do want to allocate more exposure to providers of more relevant items. We therefore restrict envy to only apply to providers with lower overall relevance.

Proportionality states that each of the n providers shall receive at least $\frac{1}{n}$ of the allocated good – in our case exposure, possibly weighted by relevance. Again, we have to take into account the variance in relevance of the different providers when defining their fractional claim of the total relevance, e.g., by weighting providers with their overall relevance.

Group Provider Fairness The task of achieving *group provider fairness* can be thought of as a multidimensional apportionment problem: Each considered provider property is a dimension along which the set of all providers is partitioned into a number of categories. Given ideal fractions for all categories of a given property, the aim is to apportion the items w.r.t. these fractions along all dimensions. These target proportions may be derived from existing proportions in a dataset, proportions of demographic groups in the real world, or set as a policy goal by external decision-makers. In IR, there are often multiple properties to consider, e.g. artists’ gender and country of origin in a music recommender system or political alignment of news outlets in a RAG system.

Multidimensional apportionment has been studied in of-line voting settings without relevance as additional parameter. Lang and Skowron focus on achieving proportional representation of the different groups individually [Lang and Skowron, 2018], while Cembrano *et al.* study the apportionment according to a true multidimensional distribution, reflecting intersectional concerns [Cembrano *et al.*, 2022]. For the latter setting, they find that the existence of solutions achieving exact proportionality is not guaranteed for more than two dimensions.

Online apportionment with changing fractional vote shares per round have also been studied, but only for a single dimension (the standard political apportionment setting) [Cembrano *et al.*, 2026]. Additionally, the selection of items in our setting is constrained by their relevance scores in each round. In IR, this distributional view has been pursued algorithmically [Balagopalan *et al.*, 2025], though without the axiomatic foundation or intersectional consideration we aim for.

Conflicting Axioms It is easy to show that strong versions of individual and group fairness axioms are not compatible. For exposure-based fairness, we can simply define provider groups such that some provider with high overall relevance has a small target fraction of exposure. A simple variant of this can be seen if we have an individual fairness axiom based on proportionality and a group fairness axiom requiring adherence to target exposure fractions. Consider a case where all providers have similar overall relevance. The proportionality axiom thus requires allocating similar fractions of relevance to all providers. However, if our group fairness target requires a specific group of providers to be allocated a large share of the exposure, it is inevitable to deviate from one of the two fairness goals. In similarly simple examples, other incompatibilities can be shown.

We therefore want to modify these axioms and introduce simplifying assumptions to the setting in order to explore combinations of individual and group fairness notions that are compatible. Simplifying assumptions include removing the uncertainty about future queries (online setting), simplifying or eliminating the relevance score, or assuming same relevance in each round. Another possible direction is to tie the individual fairness axioms to the group target proportions, i.e., to only guarantee individual fairness among providers within a certain (intersectional) group. These insights may be helpful in pinpointing the central frictions in the model.

2 Overview of Ongoing Work and Expected Results

The following roadmap summarizes our ongoing work. We currently have preliminary results for all items, and are going to further substantiate them until the workshop.

1. As a first step, we **propose suitable axiomatic properties** in this setting. This involves especially the inclusion of relevance as a key variable in fairness axioms.
2. Already simple examples show that **individual provider fairness and group provider fairness can be in conflict**. We intend to clarify the extent of these incompatibilities. Further, we will identify simplifying assumptions under which individual and group fairness can be unified.

Even strong impossibility results do not necessarily preclude algorithms that work well in practice. We are therefore running experiments in two information retrieval domains to understand fairness notions in practical settings:

3. **Recommender systems.** In collaboration with RecSys researchers, we work with music recommendation datasets. Here, provider fairness is a typical concern (i.e., fairness towards artists). Moreover, group fairness appears in the form of desiderata for recommendations regarding balancing, e.g., country of origin, language, gender, and genre.
4. **RAG:** If different sources supply data for a RAG system, they may prefer that their own material is used in question-answering (individual provider fairness) [Kim and Diaz, 2025]. In addition, sources may have different attributes, which we want to balance fairly (group provider fairness). We are currently collecting RAG data sets that allow us to test our framework in RAG settings, with special consideration of RAG-specific challenges (embedding quality, bias [Wu *et al.*, 2025; Zhang *et al.*, 2025], sensitivity to linguistic characteristics [Cao *et al.*, 2026], etc.).

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